

Question Bank - IoT Module 5

1. Define an IoT device with suitable examples.
2. List and explain the basic building blocks of an IoT device.
3. Explain the role of sensors and actuators in IoT devices.
4. Differentiate between communication and analysis & processing modules in IoT devices.
5. Describe the Raspberry Pi and its significance in IoT applications.
6. List and explain the components of a Raspberry Pi board.
7. What are the GPIO pins in the Raspberry Pi? Describe their types.
8. What operating systems are supported by Raspberry Pi?
9. Explain the features of Raspbian OS.
10. List and explain some frequently used Linux commands in the Raspberry Pi.
11. Explain the use of Serial, SPI, and I2C interfaces in Raspberry Pi.
12. What is the role of SDA and SCL pins in I2C communication?
13. Why is Python commonly used for IoT development on Raspberry Pi?
14. Write a Python program to blink an LED using Raspberry Pi GPIO.
15. Write a Python program to send an email alert when a switch is pressed.
16. Explain the working of an LDR (Light Dependent Resistor) interfaced with Raspberry Pi.
17. Compare Raspberry Pi, BeagleBone Black, and Cubieboard.
18. Write a short note on PCDuino.
19. Explain the unique features of Cubieboard in IoT applications.